Instrumental Variables (IV) Estimation

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Overview

Instrumental Variables (IV) represents one of the most powerful yet challenging methods in causal inference. When we cannot conduct a randomized experiment and face the dreaded problem of **unobserved confounding**, IV offers a potential path forward by exploiting "natural experiments" in observational data. This module introduces the theoretical foundations, assumptions, estimation methods, and practical implementation of instrumental variables, with particular emphasis on understanding what IV can and cannot identify.

Learning Objectives

By the end of this module, students will be able to:

1. **Explain** the three core assumptions of instrumental variables (relevance, exclusion restriction, exchangeability) in both technical and intuitive terms
2. **Identify** plausible instrumental variables in real-world research scenarios and critically evaluate their validity
3. **Implement** IV estimation using both manual calculations and R packages (`ivreg`, `dagitty`, `ggdag`)
4. **Interpret** IV estimates as Local Average Treatment Effects (LATE) rather than population average treatment effects (ATE)
5. **Diagnose** common IV problems including weak instruments, exclusion restriction violations, and heterogeneous treatment effects
6. **Compare** IV results with OLS estimates and explain why they differ in the presence of unobserved confounding
7. **Apply** IV methods to simulated and real datasets with appropriate caveats about external validity

R Packages

```r
pacman::p_load(
  "dagitty",
  "ggplot2",
  "ggdag",
  "tidyverse",
  "ggpubr",
  "ivreg",
  "stargazer",
  "gridExtra",
  "AER"
)

ggplot2::theme_set(ggpubr::theme_pubr())
```

1 Comprehensive Topic Explanation

The Fundamental Problem of Unobserved Confounding

In many real-world scenarios, we cannot assume that all confounders have been measured and controlled for. Traditional methods like regression adjustment, matching, or weighting fail when important confounders remain unobserved because they cannot block all backdoor paths between treatment and outcome (Hernán & Robins, 2020).

Consider these classic examples where unobserved confounding is likely:

- **Education and earnings**: Unmeasured ability affects both schooling decisions and future income
- **Hospital quality and patient outcomes**: Sicker patients may systematically choose certain hospitals
- **Treatment adherence and health**: Patient motivation influences both medication compliance and health behaviors

Formal Definition of Instrumental Variables

An **instrumental variable** Z is a variable that enables causal identification of the effect of treatment A on outcome Y by satisfying three fundamental assumptions (**neal2020introduction?**):

❗ The Three IV Assumptions

1. **Relevance**: Z is statistically associated with treatment A

$$\text{Cov}(Z, A) \neq 0$$

2. **Exclusion Restriction**: Z affects outcome Y only through its effect on treatment A

$$Z \perp\!\!\!\perp Y | A, U$$

where U represents unmeasured confounders

3. **Exchangeability**: Z is independent of all unmeasured confounders U

$$Z \perp\!\!\!\perp U$$

The IV Identification Strategy

The instrumental variables approach works by exploiting exogenous variation in treatment induced by the instrument. Rather than using all variation in treatment (which may be confounded), IV uses only the "clean" variation that comes from the instrument (**cunningham2021causalmixtape?**).

The IV estimator in the simplest case (binary instrument, binary treatment) is the Wald estimator:

$$\hat{\beta}_1^{IV} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[A|Z=1] - E[A|Z=0]}$$

This represents the ratio of the instrument's effect on the outcome (reduced form) to the instrument's effect on treatment (first stage).

What IV Identifies: Local Average Treatment Effect (LATE)

A crucial insight in IV analysis is that it typically identifies the **Local Average Treatment Effect** (LATE) rather than the population Average Treatment Effect (ATE) (Imbens & Angrist, 1994). The LATE is the average treatment effect for the subpopulation of "compliers" - those whose treatment status is affected by the instrument.

2 Intuitive Clarifications of Challenging Ideas

The "Nudge" Metaphor for Instruments

Think of an Instrument as a "Nudge"

An ideal instrument is like a gentle nudge that pushes some people toward treatment and others away from treatment, but has no other effects on their lives. It's as if nature randomly assigned some people to receive encouragement for treatment, creating a partial experiment within observational data.

The Complier Concept Through Medical Analogy

Imagine a study using copayment levels as an instrument for medication adherence:

- **Always-takers**: Patients who take medication regardless of copayment (wealthy, highly motivated)
- **Never-takers**: Patients who never take medication regardless of copayment (skeptical, non-adherent)
- **Compliers**: Patients whose medication adherence depends on copayment (price-sensitive)
- **Defiers**: Patients who do the opposite of what cost suggests (rare, usually assumed away)

IV tells us about treatment effects only for the "compliers" - those whose behavior changes with the nudge.

Why LATE ATE: The Restaurant Menu Analogy

Restaurant Menu Selection

Suppose we want to know the effect of spicy food (treatment) on satisfaction (outcome), using "chef's special" designation as an instrument:

- **Spice lovers** (always-takers): Order spicy food regardless of designation
- **Spice avoiders** (never-takers): Never order spicy food regardless
- **Marginal customers** (compliers): Order spicy food only if it's the chef's special

The effect of spicy food for marginal customers may differ from the average effect across all customers. Spice lovers might have higher tolerance and thus different satisfaction responses.

The Exclusion Restriction as "No Side Effects"

The exclusion restriction requires that the instrument affects the outcome only through the treatment, not through any other pathways. This is like requiring that a medical instrument has no side effects - it only works by influencing the treatment decision.

3 Simple, Hand-Calculation Numerical Example

College Education and Earnings

Let's study the effect of college completion on annual earnings using quarter of birth as an instrument.

Setting:

- Z: Born in Q4 (1) vs. other quarters (0)
- A: College degree (1) vs. no degree (0)
- Y: Annual earnings (in thousands of dollars)

Observed Data Summary (n = 1,000):

Birth Quarter	College Rate	Avg Earnings	Sample Size
Q1-Q3 (Z=0)	40%	$45k	750
Q4 (Z=1)	50%	$48k	250

Step-by-Step Hand Calculation

Step 1: Calculate the First Stage The first stage measures how much the instrument affects treatment probability:

$$\text{First Stage} = P(A = 1|Z = 1) - P(A = 1|Z = 0) = 0.50 - 0.40 = 0.10$$

Step 2: Calculate the Reduced Form The reduced form measures the instrument's total effect on the outcome:

$$\text{Reduced Form} = E[Y|Z = 1] - E[Y|Z = 0] = 48 - 45 = 3$$

Step 3: Calculate the IV Estimate The IV estimate is the ratio of reduced form to first stage:

$$\hat{\beta}_1^{IV} = \frac{\text{Reduced Form}}{\text{First Stage}} = \frac{3}{0.10} = 30$$

Interpretation: Among individuals whose college attendance was influenced by their birth quarter, completing college increases annual earnings by \$30,000 on average.

Understanding the Complier Population

Let's further analyze what proportion of the population are compliers:

- Overall college rate: $(750 \times 0.40 + 250 \times 0.50)/1000 = 42.5\%$
- If Q4 birth increases college attendance by 10 percentage points, then compliers represent about 10% of the population
- The \$30,000 effect applies to this 10% of the population, not necessarily to all college graduates

4 R Code Examples & Interpretation

Simulation Setup

```r
# Set Seed for Reproducibility
set.seed(123)
n <- 1000

# Generate Binary Instrument (quarter of birth)
Z <- rbinom(n, 1, 0.25) # 25% born in Q4

# Generate Unobserved Confounder (ability/motivation)
U <- rnorm(n, mean = 0, sd = 1)

# Generate Treatment (college completion)
linear_index <- -0.5 + 0.4 * Z + 0.6 * U
A <- rbinom(n, 1, plogis(linear_index))

# Generate Outcome (earnings in thousands)
Y <- 35 + 25 * A + 10 * U + rnorm(n, mean = 0, sd = 8)

# Create Data Frame
data <- data.frame(Y = Y, A = A, Z = Z, U = U)
```

DAG Visualization

```r
# Create IV DAG
iv_dag <- dagitty(
  'dag {
  Z [pos="0,0"]
  A [pos="1,0"]
  Y [pos="2,0"]
  U [pos="1,1"]

  Z -> A
  A -> Y
  U -> A
  U -> Y
}'
)

# Plot Using Ggdag
ggdag(iv_dag) +
  theme_dag()
```

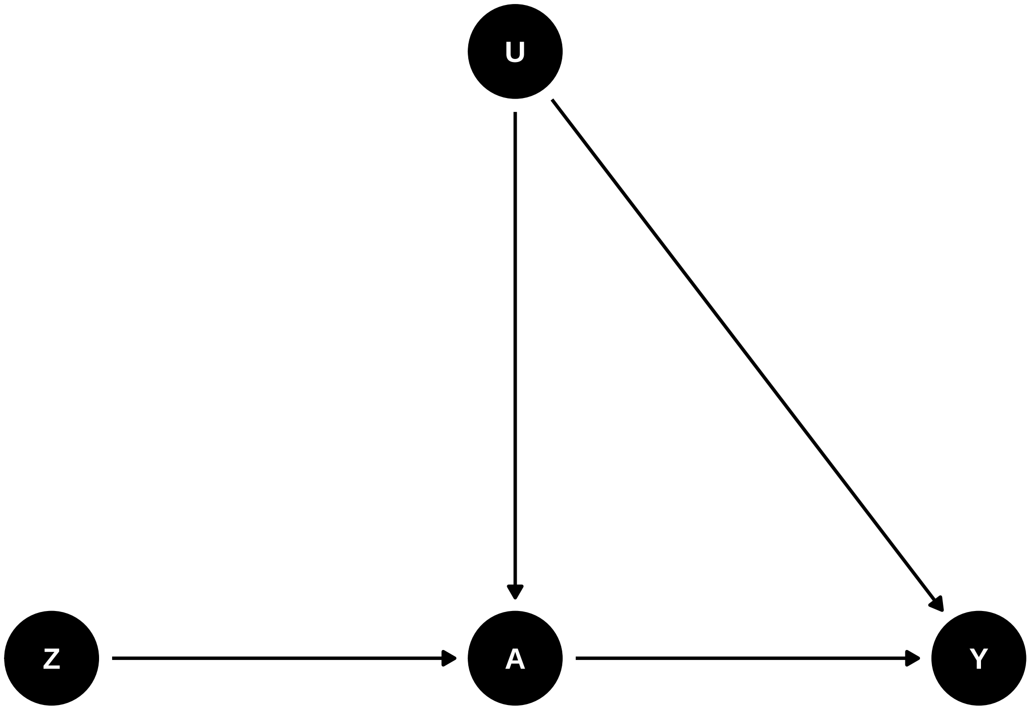


Figure 1: Directed Acyclic Graph for IV Setting

Manual IV Calculation

```
# Manual Calculation of IV Estimate
first_stage <- mean(data$A[data$Z == 1]) - mean(data$A[data$Z == 0])
reduced_form <- mean(data$Y[data$Z == 1]) - mean(data$Y[data$Z == 0])
iv_estimate <- reduced_form / first_stage

cat("First Stage:", round(first_stage, 3), "\n")
```

```
First Stage: 0.093
```

```
cat("Reduced Form:", round(reduced_form, 3), "\n")
```

```
Reduced Form: 2.492
```

```r
cat("Manual IV Estimate:", round(iv_estimate, 3), "\n")
```

```
Manual IV Estimate: 26.798
```

```r
# Compare with True Effect and OLS
true_effect <- 25
ols_estimate <- coef(lm(Y ~ A, data = data))["A"]

cat("True Causal Effect:", true_effect, "\n")
```

```
True Causal Effect: 25
```

```r
cat("OLS Estimate (biased):", round(ols_estimate, 3), "\n")
```

```
OLS Estimate (biased): 31.189
```

```r
cat("IV Estimate:", round(iv_estimate, 3), "\n")
```

```
IV Estimate: 26.798
```

Formal IV Estimation

```r
# Using Ivreg Package for IV Estimation
iv_model <- ivreg(Y ~ A | Z, data = data)
summary(iv_model)
```

```

Call:
ivreg(formula = Y ~ A | Z, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-38.0331  -8.9414  -0.1814   8.5961  45.1099
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   34.200      4.214   8.116  1.4e-15 ***
A             26.798      9.940   2.696  0.00714 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.62 on 998 degrees of freedom
Multiple R-Squared: 0.5939, Adjusted R-squared: 0.5935
Wald test: 7.268 on 1 and 998 DF,  p-value: 0.007136
```

```r
# Compare with OLS
ols_model <- lm(Y ~ A, data = data)

# Create Comparison Table
stargazer(
  ols_model,
  iv_model,
  type = "text",
  column.labels = c("OLS (Biased)", "IV (Causal)"),
  omit.stat = c("adj.rsq", "f")
)
```

```
===========================================================
                               Dependent variable:
                          ----------------------------
                                         Y
                              OLS         instrumental
                                            variable
                          OLS (Biased)   IV (Causal)
                              (1)            (2)
-----------------------------------------------------------
A                           31.189***      26.798***
                            (0.796)        (9.940)


Constant                    32.347***      34.200***
                            (0.517)        (4.214)
```

```
-----------------------------------------------------------
Observations                    1,000          1,000
R2                              0.606          0.594
Residual Std. Error (df = 998)  12.436         12.624
===========================================================
Note:                          *p<0.1; **p<0.05; ***p<0.01
```

Instrument Strength Check

```r
# Check First Stage Strength
first_stage_reg <- lm(A ~ Z, data = data)
first_stage_f <- summary(first_stage_reg)$fstatistic[1]

cat("First Stage F-statistic:", round(first_stage_f, 2), "\n")
```

```
First Stage F-statistic: 6.64
```

```r
cat("Strong instrument?", first_stage_f > 10, "\n")
```

```
Strong instrument? FALSE
```

5 Hands-On R Lab

Lab Objectives

In this lab, you will: 1. Simulate data with unobserved confounding and estimate treatment effects using IV 2. Practice implementing IV estimation manually and with built-in functions 3. Diagnose instrument strength and understand its importance 4. Explore how different instruments identify different LATEs 5. Reflect on the external validity of IV estimates

Pre-Lab Reading Warm-up

Before starting this lab, review: - Chapter 16 of Hernán & Robins (2020) on instrumental variables - The three core IV assumptions and their intuitive meaning - The difference between LATE and ATE

Data Simulation

```r
# Simulate Data for Lab Exercises
set.seed(579)
n_lab <- 1500

# Instrument 1: Distance to College (miles)
distance <- runif(n_lab, 0, 50)

# Instrument 2: Tuition Subsidy Eligibility (binary)
subsidy <- rbinom(n_lab, 1, 0.3)

# Unobserved Ability
ability <- rnorm(n_lab, 0, 2)

# Education (affected by Instruments and ability)
education <- 12 +
  (-0.05) * distance +
  0.8 * subsidy +
  0.7 * ability +
  rnorm(n_lab, 0, 1.5)
education <- pmax(education, 8) # Minimum 8 years

# Earnings (affected by Education and ability)
earnings <- 2.2 + 0.07 * education + 0.15 * ability + rnorm(n_lab, 0, 0.4)

lab_data <- data.frame(
  log_earnings = earnings,
  education = education,
  distance = distance,
  subsidy = subsidy,
  ability = ability # Note: in real data, we wouldn't observe this
)
```

Lab Tasks

Task 1: Exploratory Analysis

```r
# Task 1: Examine the Relationships between Variables
# 1.1 Check the Correlation between Distance and Education
cor(lab_data$distance, lab_data$education)

# 1.2 Check the Correlation between Subsidy and Education
cor(lab_data$subsidy, lab_data$education)

# 1.3 Create Scatter Plots of Education Vs Earnings, and Distance Vs Education
# YOUR CODE HERE
```

```r
# Solution for Task 1
cat(
  "Correlation between distance and education:",
  round(cor(lab_data$distance, lab_data$education), 3),
  "\n"
)
```

```
Correlation between distance and education: -0.321
```

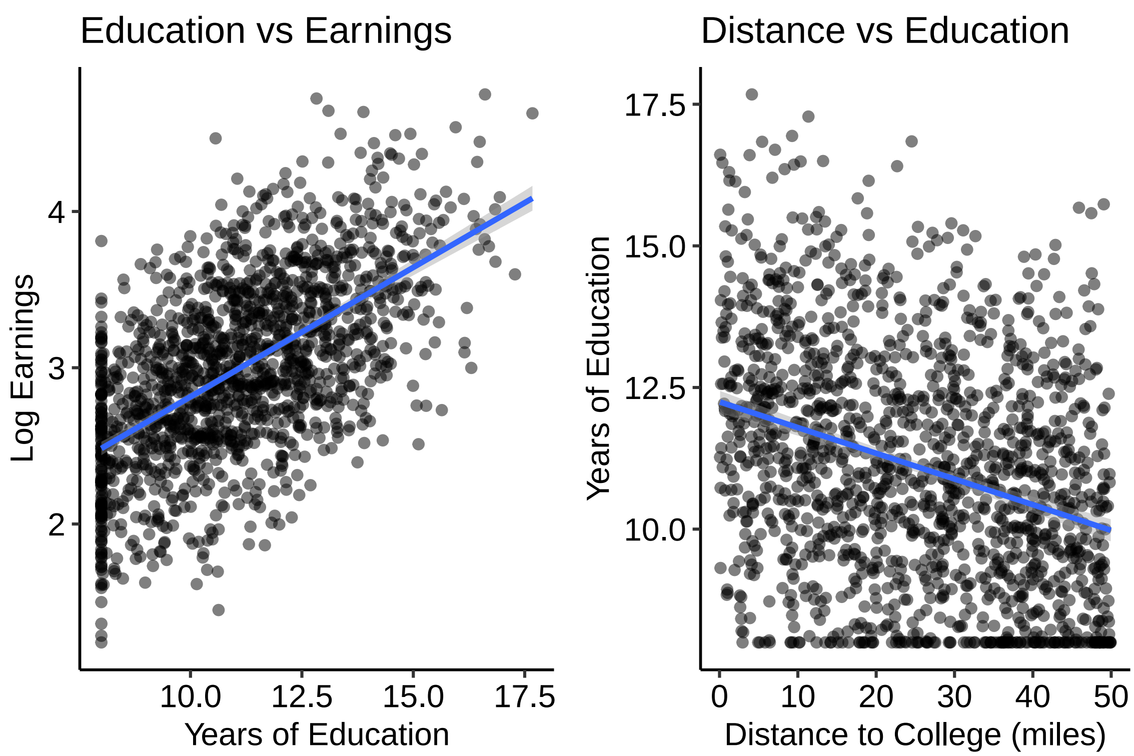
```r
cat(
  "Correlation between subsidy and education:",
  round(cor(lab_data$subsidy, lab_data$education), 3),
  "\n"
)
```

```
Correlation between subsidy and education: 0.158
```

```r
# Scatter Plots
p1 <- ggplot(lab_data, aes(x = education, y = log_earnings)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  labs(
    title = "Education vs Earnings",
    x = "Years of Education",
    y = "Log Earnings"
  )
```

```r
p2 <- ggplot(lab_data, aes(x = distance, y = education)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  labs(
    title = "Distance vs Education",
    x = "Distance to College (miles)",
    y = "Years of Education"
  )

grid.arrange(p1, p2, ncol = 2)
```



Task 2: OLS Estimation

```r
# Task 2: Estimate the Education Effect Using OLS
# 2.1 Run OLS Regression of Earnings on Education
# ols_model <- YOUR CODE HERE

# 2.2 Interpret the Coefficient on Education
# The OLS Estimate Suggests That ...
```

```r
# Solution for Task 2
ols_model <- lm(log_earnings ~ education, data = lab_data)
summary(ols_model)
```

```

Call:
lm(formula = log_earnings ~ education, data = lab_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.46922 -0.31279 -0.00106  0.32954  1.56016 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 1.159182   0.066444   17.45   <2e-16 ***
education   0.165524   0.005881   28.14   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4676 on 1498 degrees of freedom
Multiple R-squared:  0.3459,	Adjusted R-squared:  0.3454 
F-statistic: 792.1 on 1 and 1498 DF,  p-value: < 2.2e-16
```

```r
cat(
  "OLS estimate of education effect:",
  round(coef(ols_model)["education"], 4),
  "\n"
)
```

```
OLS estimate of education effect: 0.1655 
```

```r
cat(
  "Interpretation: Each additional year of education is associated with a",
  round(exp(coef(ols_model)["education"]) - 1, 4) * 100,
  "% increase in earnings, but this may be biased by unobserved ability."
)
```

```
Interpretation: Each additional year of education is associated with a 18 % increase in earnings, but thi
```

Task 3: IV Estimation with Distance Instrument

```
# Task 3: Use Distance as an Instrument for Education
# 3.1 Estimate First Stage Regression
# first_stage_distance <- YOUR CODE HERE

# 3.2 Check Instrument Strength (F-statistic)
# F-statistic =

# 3.3 Estimate IV Regression Using Distance Instrument
# iv_distance <- YOUR CODE HERE

# 3.4 Compare with OLS Estimate
```

```
# Solution for Task 3
first_stage_distance <- lm(education ~ distance, data = lab_data)
f_stat_distance <- summary(first_stage_distance)$fstatistic[1]

iv_distance <- ivreg(log_earnings ~ education | distance, data = lab_data)

cat("First stage F-statistic:", round(f_stat_distance, 2), "\n")
```

```
First stage F-statistic: 172.29
```

```
cat("Strong instrument?", f_stat_distance > 10, "\n")
```

```
Strong instrument? TRUE
```

```
cat(
  "IV estimate using distance:",
  round(coef(iv_distance)["education"], 4),
  "\n"
)
```

```
IV estimate using distance: 0.0537
```

```r
cat("OLS estimate:", round(coef(ols_model)["education"], 4), "\n")
```

```
OLS estimate: 0.1655
```

Task 4: IV Estimation with Subsidy Instrument

```r
# Task 4: Use Subsidy as an Instrument for Education
# 4.1 Estimate First Stage with Subsidy
# first_stage_subsidy <- YOUR CODE HERE

# 4.2 Check Instrument Strength
# F-statistic =

# 4.3 Estimate IV Regression Using Subsidy
# iv_subsidy <- YOUR CODE HERE

# 4.4 Compare with Distance Instrument Results
```

```r
# Solution for Task 4
first_stage_subsidy <- lm(education ~ subsidy, data = lab_data)
f_stat_subsidy <- summary(first_stage_subsidy)$fstatistic[1]

iv_subsidy <- ivreg(log_earnings ~ education | subsidy, data = lab_data)

cat("First stage F-statistic (subsidy):", round(f_stat_subsidy, 2), "\n")
```

```
First stage F-statistic (subsidy): 38.28
```

```r
cat("Strong instrument?", f_stat_subsidy > 10, "\n")
```

```
Strong instrument? TRUE
```

```r
cat("IV estimate using subsidy:", round(coef(iv_subsidy)["education"], 4), "\n")
```

```
IV estimate using subsidy: 0.0676
```

```r
cat(
  "IV estimate using distance:",
  round(coef(iv_distance)["education"], 4),
  "\n"
)
```

```
IV estimate using distance: 0.0537
```

Task 5: Multiple Instruments

```r
# Task 5: Use both Instruments together
# 5.1 Estimate IV with both Instruments
# iv_both <- YOUR CODE HERE

# 5.2 Perform Overidentification Test
# YOUR CODE HERE

# 5.3 Interpret the Results
```

```r
# Solution for Task 5
iv_both <- ivreg(log_earnings ~ education | distance + subsidy, data = lab_data)

# Overidentification Test
library(AER)
overid_test <- summary(iv_both, diagnostics = TRUE)
cat(
  "Overidentification test p-value:",
  round(overid_test$diagnostics["Sargan", "p-value"], 4),
  "\n"
)
```

```
Overidentification test p-value: 0.7655
```

```r
cat(
  "IV estimate with both instruments:",
  round(coef(iv_both)["education"], 4),
  "\n"
)
```

```
IV estimate with both instruments: 0.0565
```

```r
cat(
  "Do instruments identify the same parameter?",
  overid_test$diagnostics["Sargan", "p-value"] > 0.05,
  "\n"
)
```

```
Do instruments identify the same parameter? TRUE
```

Lab Reflection Questions

1. **Retrieval Quiz**: What are the three core assumptions of instrumental variables, and why is each important?
2. **Critical Thinking**: Why did the two different instruments give slightly different estimates? What does this tell us about the complier populations?
3. **Application**: If you were designing a policy to increase educational attainment, which IV estimate would be more relevant and why?
4. **Limitations**: What are the main threats to the validity of each instrument in this example?

6 Common Pitfalls & Checks

⚠ Weak Instruments

Weak instruments (first stage F-statistic < 10) can cause: - Substantial bias in IV estimates, often in the direction of OLS bias - Inflated Type I error rates - Unreliable inference
Solution: Always report first stage F-statistic and consider weak instrument robust methods.

⚠ Exclusion Restriction Violations

If the instrument affects the outcome through pathways other than the treatment, IV estimates are biased. Common violations include: - Direct effects of the instrument on the outcome - Effects through mediating variables not captured by the treatment
Solution: Use domain knowledge to argue for exclusion, test with multiple instruments.

Interpretation as LATE Vs ATE

IV typically identifies the Local Average Treatment Effect (LATE) for compliers, not the population Average Treatment Effect (ATE).
Solution: Clearly describe the complier population and discuss external validity limitations.

! Essential Diagnostic Checks

1. **First stage strength**: F-statistic > 10
2. **Overidentification tests**: With multiple instruments, check if they identify the same parameter
3. **Comparison with OLS**: Large differences may indicate substantial selection bias
4. **Sensitivity analysis**: Test robustness to exclusion restriction violations

7 Advanced Teaching Activities & Interactivity

Think-Pair-Share: Instrument Validity

Prompt: "Consider using rainfall as an instrument for agricultural productivity in a study of how economic conditions affect civil conflict. Pair up and discuss: (1) Is rainfall a relevant instrument? (2) Does it satisfy the exclusion restriction? (3) Is it exogenous? Be prepared to share your reasoning."

Flipped Classroom Exercise

Pre-class assignment: Read (**angrist1991does?**) on using quarter of birth as an instrument for education. Come to class prepared to discuss: "What population does this instrument identify effects for, and how might this limit the policy relevance of the findings?"

Problem-Based Case Study

Scenario: A hospital wants to evaluate the effect of a new surgical technique on patient outcomes. They're considering using surgeon preference as an instrument, since some surgeons adopted the technique earlier than others.

Tasks: 1. Design the IV analysis plan 2. Identify potential threats to each assumption 3. Propose sensitivity analyses 4. Discuss how you would communicate results to hospital administrators

Spaced Retrieval Quiz

1. **Week 1**: What are the three core IV assumptions?
2. **Week 3**: How do we calculate the first stage F-statistic and what does it tell us?
3. **Week 6**: What is the difference between LATE and ATE, and why does it matter?
4. **Week 8**: How can multiple instruments help test the exclusion restriction?

Self-Explanation Task

"Explain in your own words why IV estimates can be more biased than OLS when instruments are weak. Use the formula for the IV estimator in terms of the first stage to structure your explanation."

8 Appendix: Advanced Theory & Derivations

Potential Outcomes Framework for IV

Let $Y_i(a, z)$ be the potential outcome for individual i under treatment a and instrument z, and $A_i(z)$ be the potential treatment under instrument z.

The observed data are: - Z_i: Instrument - $A_i = A_i(Z_i)$: Observed treatment - $Y_i = Y_i(A_i(Z_i), Z_i)$: Observed outcome

Principal Stratification

Based on potential treatment responses $(A_i(0), A_i(1))$, individuals fall into four principal strata:

1. **Compliers**: $A_i(0) = 0, A_i(1) = 1$
2. **Always-takers**: $A_i(0) = 1, A_i(1) = 1$
3. **Never-takers**: $A_i(0) = 0, A_i(1) = 0$
4. **Defiers**: $A_i(0) = 1, A_i(1) = 0$ (ruled out by monotonicity)

LATE Theorem

Under the IV assumptions plus monotonicity, the Wald estimator identifies:

$$\frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[A|Z = 1] - E[A|Z = 0]} = E[Y(1) - Y(0)|\text{Complier}]$$

This is the Local Average Treatment Effect for compliers (Imbens & Angrist, 1994).

Efficiency Bounds

The semiparametric efficiency bound for IV estimation under homoskedasticity is:

$$V = \frac{\sigma^2}{n \cdot \text{Var}(Z) \cdot R^2_{A|Z}}$$

where $R^2_{A|Z}$ is the first stage R-squared. This shows why weak instruments lead to imprecise estimates.

References

Hernán, M. Á., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.

Imbens, G. W., & Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica : Journal of the Econometric Society*, *62*(2), 467–475.